

Faithful two-pole closure for the viscous drop modes

Reid's arbitrary-Oh eigenvalues mapped onto the drop-mode ODE

SpectralKM derivations

1 What is being matched, and why

Each droplet surface mode in this model is carried by a second-order ODE (`eq:lambda-1`; implemented in `src/affine.jl:drop_affine`)

$$\ddot{\beta}_l + 2\lambda_l \dot{\beta}_l + \omega_l^2 \beta_l = F_l b_l, \quad (1)$$

whose characteristic polynomial is $s^2 + 2\lambda_l s + \omega_l^2$. The published choice takes Lamb's (1881) small-viscosity asymptotics,

$$\lambda_l^{\text{Lamb}} = \text{Oh} (l-1)(2l+1), \quad (\omega_l^{\text{Lamb}})^2 = \omega_{l,0}^2 = l(l-1)(l+2), \quad (2)$$

with $\omega_{l,0}$ the inviscid Rayleigh–Lamb capillary frequency. Reid (1960) instead gives the *exact* linear viscous eigenvalues at arbitrary Oh, as roots σ of a transcendental characteristic equation (see `src/reid.jl`); with σ a decay rate, perturbations behave as $e^{-\sigma t}$, so the ODE roots are $s = -\sigma$.

This note fixes the remaining question: *given a pair of Reid eigenvalues, what are the faithful (λ_l, ω_l^2) ?* The algebra is verified symbolically in `derivations/cas_reid_two_pole.py` (all assertions pass).

2 The mapping is Vieta, and needs no case split

For a monic quadratic with roots s_1, s_2 ,

$$(s - s_1)(s - s_2) = s^2 + 2\lambda_l s + \omega_l^2 \iff \boxed{\lambda_l = -\frac{1}{2}(s_1 + s_2), \quad \omega_l^2 = s_1 s_2} \quad (3)$$

exactly, for *any* pair. Two consequences.

Underdamped ($s = -\gamma \pm i\omega$).

$$\lambda_l = \gamma, \quad \omega_l^2 = \gamma^2 + \omega^2, \quad (4)$$

not ω^2 . The distinction matters because (1) oscillates at $\sqrt{\omega_l^2 - \lambda_l^2}$, and only (4) returns ω . Lamb (2) carries the same $O(\lambda^2/\omega^2)$ slip, but there it sits inside its own asymptotic error; here it would be gratuitous.

Overdamped ($s = -\gamma_1, -\gamma_2$, **real**, $\gamma_1 < \gamma_2$).

$$\lambda_l = \frac{1}{2}(\gamma_1 + \gamma_2), \quad \omega_l^2 = \gamma_1 \gamma_2, \quad \lambda_l^2 - \omega_l^2 = \left(\frac{\gamma_1 - \gamma_2}{2}\right)^2 \geq 0, \quad (5)$$

so the ODE is genuinely overdamped, degenerating only when the rates coincide. As $\gamma_2 \rightarrow \gamma_1$ this reduces continuously to the critically damped case, hence (3) is a strict generalisation.

3 The superseded choice, and why it was badly wrong

An earlier revision of `drop_viscous_coeffs` set, for a real root,

$$\lambda_l = \gamma_{\text{tracked}}, \quad \omega_l^2 = \lambda_l^2, \quad (6)$$

a double root at the tracked rate. This discards γ_2 and invents a second rate. Two compounding errors made it worse than a mild approximation:

1. **Wrong branch.** Continuation in Oh from the complex pair does *not* land on the slowest real root once the pair has merged. Measured at $l = 2$ by scanning the real axis (the residual is real for real $s > 0$, since $q = \sqrt{s/\text{Oh}}$ is then real and so is j_l):

Oh	real roots on $(10^{-4}, 200)$	tracked	slowest
0.8	2.040, 3.670, 23.94, 63.04, 118.3, 189.4	3.670	2.040
1.0	1.280, 5.846, 29.93, 78.80, 147.9	5.846	1.280
3.0	0.357, 20.97, 89.86	20.97	0.357

Continuation consistently returns the *second* root. The slowest root decreases with Oh (as a very viscous drop must relax ever more slowly, rate $\propto 1/\text{Oh}$), whereas the tracked one grows — the diagnostic that exposed this.

2. **Wrong stiffness.** Applying (3) to the two slowest roots gives $\omega_l^2 = 7.487, 7.485, 7.480$ at Oh = 0.8, 1.0, 3.0, against $\omega_{2,0}^2 = l(l-1)(l+2) = 8$. The faithful mapping therefore **preserves the capillary stiffness**, essentially independent of Oh, as it must: viscosity damps, it does not change what restores. The superseded choice (6) instead gave $\omega_l^2 = \lambda_l^2 = 20.97^2 \approx 440$ at Oh = 3, a stiffness too large by a factor 59.

That $\omega_l^2 \approx \omega_{l,0}^2$ is not an input to the construction; it emerges, and so serves as a cheap correctness check on the root pair.

4 Where the overdamped branch is actually reached

The capillary pair merges onto the real axis (mode becomes aperiodic) at an Oh that falls with l : measured between 0.5 and 0.8 for $l = 2, 4$, and between 0.3 and 0.5 for $l = 8, 16$. So the overdamped case is not hypothetical for Oh $\gtrsim 0.5$, and high modes enter it first.

5 Algorithm

For each l , at the requested Oh:

1. Track the least-damped eigenvalue by continuation in Oh from a value small enough that Lamb is genuinely asymptotic (`reid_root_tracked`).
2. If $|\text{Im } \sigma| > 0$: the pair is $\sigma, \bar{\sigma}$; apply (4).
3. If $\text{Im } \sigma = 0$: the pair has merged. Find the two real roots by a *bracketed* search, not a scan — see §6 — and apply (5). Do *not* rely on the continued root being the slowest.
4. Sanity-check ω_l^2 against $\omega_{l,0}^2$; a large discrepancy indicates a bad root pair rather than physics.

6 Bracketing the dominant pair

The real roots need not be hunted for. In

$$\text{rhs} = \frac{2(l-1)}{q^2} \left[l + (l+1) \frac{q-2lQ}{q-2Q} \right] \quad (7)$$

the poles of Q (the zeros of j_l) are **removable**, since $(q-2lQ)/(q-2Q) \rightarrow l$ as $Q \rightarrow \infty$. The true singularities are the zeros of $g(q) = q - 2Q(q)$, and the characteristic roots *interlace* with them: measured at $l = 2$, $\text{Oh} = 0.8$, the singularities sit at $s = 22.56, 62.72, 118.13, 189.26$ and the roots at $23.94, 63.04, 118.31, 189.37$ — each root just above its own singularity.

The dominant pair lies below the *first* singularity, so the entire search is confined to

$$0 < s < s^*(l, \text{Oh}) = \text{Oh } q^*(l)^2, \quad (8)$$

with $q^*(l)$ the first zero of g . That zero is uniquely bracketed: $g > 0$ as $q \rightarrow 0^+$ (since $Q \sim q/(2l+3)$ gives $g \sim q[1 - 2/(2l+3)]$), $g \rightarrow -\infty$ at the first zero jz_1 of j_l , and g has a single maximum between — so bisection on $(0, jz_1)$ is rigorous. Crucially q^* depends on l *alone*, not on Oh , so it is computed once and memoised; measured $q^*/l = 2.656, 1.972, 1.577, 1.346, 1.209, 1.134, 1.082, 1.051$ for $l = 2, \dots, 240$, always just below jz_1 .

Verified over $l = 2 \dots 32$ and $\text{Oh} = 0.05 \dots 3$ (30 combinations): the interval (8) contains **exactly** 0 real roots while the mode oscillates and **exactly** 2 once the pair has merged — never 1 or 3. That parity is asserted in `test/test_reid.jl`; an odd count would mean the dominant pair has been mis-identified and every downstream damping rate is wrong.

A superseded revision instead scanned 40 000 log-spaced points over a guessed range. Anchor jz_1 to a bisection of `besselj` rather than to an asymptotic estimate: a McMahon first-zero formula overshoots by 0.016 at $l = 2$, which places the bracket on the far side of the pole and recovers the *second* singularity (8.854) in place of the first (5.311).

7 Evaluating the Bessel ratio, and whether the seed truncation is justified

The characteristic equation needs only the ratio $Q_l(q) = j_{l+1}(q)/j_l(q)$, never j_l itself. Forming j_l is in fact unsafe: `besselj` *overflows* at small Oh , where $|q| \sim 50$ with a large imaginary part, and *underflows* at large order with small argument. The latter is not hypothetical — during development it silently returned $q^*(120) = 0.271$ against a true 129.86, precisely the class of failure that yields a plausible-looking wrong answer rather than an error.

We therefore use the continued fraction in recurrence form. From $j_{n-1} + j_{n+1} = \frac{2n+1}{q} j_n$, dividing by j_n ,

$$Q_{n-1} = \frac{1}{\frac{2n+1}{q} - Q_n}, \quad Q_{n_0} \simeq \frac{q}{2n_0 + 3}, \quad (9)$$

recursed downward from n_0 to l .

The seed is *not* justified by being accurate. Expanding the exact ratio (verified in `derivations/cas_bessel_`

$$Q_n = \frac{q}{2n+3} \left[1 + \frac{q^2}{4n^2 + 16n + 15} + O(q^4) \right], \quad (10)$$

so the seed's *relative* remainder is $O(q^2/n^2)$, which is $O(1)$ — not small — whenever n_0 is merely comparable to $|q|$. Taking (9)'s seed on its own would be an unjustified truncation.

It is justified because the recurrence contracts. Differentiating (9),

$$\frac{\partial Q_{n-1}}{\partial Q_n} = Q_{n-1}^2 \implies e_{n-1} = Q_{n-1}^2 e_n \implies E_{n-1} = Q_{n-1} Q_n E_n \quad (11)$$

for absolute error e and relative error $E = e/Q$. With $Q_n \sim q/(2n+3)$ the per-step relative factor is $q^2/[(2n+1)(2n+3)] \sim (q/2n)^2 \ll 1$ as soon as $2n \gg |q|$, and the cumulative factor from n_0 down to $l+1$ is a product of such terms — geometric, hence accuracy exponential in the padding. This is the Miller premise.

Consequence for the testbed. Since the justification is contraction rather than seed accuracy, the property to assert in the source is **pad-independence**, not agreement of the seed with anything. Measured relative error against a scaled-Bessel reference:

l	q	pad 0	2	8	30	120
2	5	2.7×10^{-4}	1.1×10^{-6}	1.6×10^{-15}	1.2×10^{-16}	1.2×10^{-16}
2	$37.6 + 37.6i$	1.2×10^{-16}		(flat)		
16	21.5	4.8×10^{-14}	1.1×10^{-15}		(flat)	

There is a genuinely failing regime at small padding, so the test discriminates. The shipped choice $n_0 = l + \max(60, l/2 + |q|) + |q|$ sits in the plateau at every point checked (l up to 240), with doubling the padding changing no digit.

8 Scope

Two limitations are inherited and unchanged by this note. First, (1) is a two-pole model: Reid’s spectrum also contains an infinite diffusive ladder (internal Stokes relaxations, rates $\approx \text{Oh } q_j^2$ with q_j the zeros of j_l), so the *forced* response carries memory that no second-order ODE reproduces. Matching two poles exactly is right for free decay and approximate under forcing; the n -pole generalisation is deferred. Second, the residues are not used here — only the poles — so this note says nothing about the amplitude with which pressure forcing excites each mode.